2.2.2 SCA Window Mode Measurements

- Switch the SCA to *window* mode
- NOTE: For window mode, an upper level setting of 10 corresponds to a 1.0V window width
- Select a reasonable LLD setting
- Vary the window width and record the minimum and maximum pulser amplitudes that will produce a SCA output pulse for a given window setting (Determine the full range of input pulse heights that will produce a SCA output pulse).
- Repeat for two more different LLD settings

2.2.3 SCA Normal Mode Measurements

This exercise is to help you determine the relationship between the upper level discriminator (ULD) and LLD dial settings.

- Set the SCA to *normal* mode
- For various LLD and ULD settings find the minimum and maximum pulse amplitudes that result in logic pulse output from the SCA.

3 Personal Computer Operation and the Multichannel Analyzer (MCA)

The multichannel analyzers you will use throughout the remainder of the term are installed in an expansion slot of you lab station computer. With the aid of the instructor, familiarize yourself with ORTEC Maestro, the software used to produce and display a multichannel spectrum. The input pulses from the amplifier to the MCA must be positive in polarity between 0-10 V.

3.1 Equipment Setup

3.1.1 Measurement Setup

- Connect the pulser to the amplifier
- Connect the output of the amplifier to the input of the MCA (on the back of your computer)
- Set the pulser amplitude and amplifier gain to produce pulses that meet the MCA input requirements
- NOTE: Since the pulser amplitude is fixed, all pulses should be stored in a single channel

*What channel is associated with the pulser amplitude?
*How many channels does your MCA have?

3.2 MCA Measurements

3.2.1 MCA General Measurements

- Set the pulser frequency to about 1 kHz
- Split the signal from the amplifier, directing one branch into the oscilloscope and the other into the MCA
- Record the MCA channel and pulse amplitude while varying the pulser amplitude. Do not erase the MCA after each step. The result is a series of spikes that mark the channel corresponding to each discrete pulser amplitude.
- Record the net count under each peak measured for the same live time.
- Change the amplifier gain, and repeat the measurement.

Example table for recording data

Pulser amplitude	Amplifer gain	Amplifer amplitude	Channel number	Channel counts
Amp 1	gain 1			
Amp 2	gain 1			
...Amp 5	gain 1			
Amp 1	gain 2			
Amp 2	gain 2			
...Amp 5	gain 2			
Amp 1	gain 3			
Amp 2	gain 3			